

An Integrated Deep Learning Framework Approach for Nail Disease Identification

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Abstract— Nail Diseases refer to some kind of deformity in the nail unit. Although the nail unit is a skin accessory, it has its own distinct class of diseases as these diseases have their own set of signs, symptoms, causes and effects that may or may not relate to other medical conditions. Recognizing nail diseases still remains an unexplored and a challenging endeavor in itself. This paper proposes a novel deep learning framework to detect and classify nail diseases from images. A distinct class of eleven diseases i.e. onychomycosis, subungual hematoma, beau's lines, yellow nail syndrome, psoriasis, hyperpigmentation, koilonychias, paronychia, pincer nails, leukonychia, and onychorrhexis. The framework uses a hybrid of Convolutional Neural Network (CNNs) for feature extraction. Due to the non-existence of a meticulous dataset, a new dataset was built for testing the enactment of our proposed framework. This work has been tested on our dataset and has also been compared with other state-of-the-art algorithms (SVM, ANN, KNN, and RF) that have been shown to have an excelled performance in the area of feature extraction. The results have shown a comparable performance, in terms of differentiating amongst the wide spectrum of nail diseases and are able to recognize them with an accuracy of 84.58%.

Keywords: Deep Convolutional Neural Networks, Deep Learning, Nail Diseases, Feature Extraction.

I. INTRODUCTION

The Nail, made up of a hardened protein called Keratin, is the unit that protects the tissues of the toes and fingers. The condition of the nail unit can serve as a direct indicator to one's health. Specific types of nail discolorations, hyperpigmentation, malignancies, changes in the nail

texture, and appearances of some specific conditions hint towards diseases such as diabetes, psoriatic arthritis, anaemia, fungal infections, melanoma, liver diseases and malnutrition.

A person might get a nail disease at any age [1]. Approximately, 15% of the overall population suffers from onychomycosis, a fungal infection which can be quite painful. Most of the nail diseases are of infectious origin. Some life threatening diseases like subungual melanoma, which has an incubation period of approximately 2 years and a 10 year survival rate of 43%, are not easy to detect in their early stages. Till 2003, no significant work had been done for the detection of nail abnormalities, [2]. Nail Disease detection has begun to gain importance now, many researchers have started to show interest in the field. The Nail unit is used to predict [3] and detect psoriatic arthritis, texture and colour analysis for disease detection [4], early disease detection and related work has been done.

Dermatology is a field of medical science that has begun to catch the attention of the researchers. Image matching algorithms [5], automated melanoma screening, skin cancer detection, categorization of dermatological images, and dermatological disease detection are some of the newly pursued areas. Deep Learning, which uses Artificial Neural Networks that contain more than one layer, has been an influential topic and keeps on attracting many researchers to try their hands on it. Esteva et al, [6], have used deep neural networks for classification of skin cancer.

Techniques of deep learning have given tremendously improved results as compared to the conventional models like Support Vector Machine (SVM), K Nearest Neighbourhood (KNN) and Random Forest (RF). It has been shown that neural networks such as these have the ability of approximating any function arbitrarily well [7]. Furthermore, it has been shown that adding more layers improve the performance of the network faster than all other

improvements [8].

Deep learning techniques use computational models that contain multiple processing layers with levels of abstraction. Such techniques have produced better results as compared to the usual methods in visual object recognition and detection.[9] Deep learning detects sophisticated structures in large data sets by using the back-propagation algorithm to indicate how a machine should change its internal parameters that are used to compute the representation in each layer from the representation in the previous layer. Deep convolutional nets have brought about breakthroughs in processing images and video using such techniques [10].

Various deep learning models were created to perform dissimilar tasks. In the case of visual computing, the algorithms were created based on neuroscience with the intention of approximating the function and behaviour of the human visual cortex (Neural Networks). These models imitate the neurons of the brain, which fulfil different levels of abstraction, and combine the characteristics of previous levels with the current ones. These models are known as Convolutional Neural Networks. The architecture of CNN has more than one deep layer with different levels of abstraction. [11].

Recently, Convolutional Neural Networks have gained popularity in feature learning and object classification, considering the fact that the use of high performance GPU makes training a large dataset possible which leads to a better performance and increased accuracy [12].

The domain of nail disease identification is still unpursued. Although the nail unit has been analysed and detected for various purposes like detection of specific diseases such as melanoma and psoriatic arthritis, not much light has been shed on the identification of nail diseases., Our approach exploits CNN to detect and classify a distinctive class of eleven nail diseases, namely: onychomycosis, subungual hematoma, beau's lines, yellow nail syndrome, psoriasis, hyperpigmentation, koilonychia, paronychia, pincer nails, leukonychia, and onychorrhexis.

The challenges faced by the problem statement include: unavailability of good quality datasets for use, the shape and texture of the nail varies from individual to individual, so we need an efficient and accurate means of feature extraction.

Keeping the above challenges in mind, in this paper, we have proposed a new deep learning framework that uses an ensemble of Convolutional Neural Nets.

We have applied a hybrid of three separate Convolutional

Neural networks. The feature vectors obtained from the same are combined together on which the Random Forest Classifier is applied to classify the aforementioned class of diseases. Accuracy assessment is performed in which accuracy, and, kappa coefficients are computed along with ROC curves.

The paper is ordered as follows: Introduction (section 1), Literature Review (section 2), Data Collection (section 3), Methodology—classifiers, proposed approach (section 4), Results (section 5) and finally, Conclusion (section 6).

II. LITERATURE REVIEW

Recently, a lot of researchers have been showing interest in detection of diseases using computer vision techniques such as feature extraction, and classification. Human Nail Image Processing is a newly pursued area in which the human nail image has been used to identify specific types of diseases, as mentioned above.

Trupti et al [13] have used image processing technique on the human nail unit to create an early stage disease detection system. To be specific, the feature used is the nail pigment that gave an accuracy of 65%. Liao, Haofu [14] have devised a deep learning approach for universal skin disease classification, using 23 top level categories of the Dermnet taxonomy, with Nail Diseases being one of the category with very limited images per class for nail diseases. Another distinguished usage of the nail unit has been seen in computer aided Nailfold capillaroscopy image analysis, performed by Niraj and Gerald [15], in which they have performed an enhancement on the images obtained from Nail Capillaroscopy; this method is used for assessment of the condition of the blood capillaries of the Nailfold which finds its application in detection of diseases such as scleroderma, and Raynaud's phenomenon. A colour histogram approach for the analysis of Psoriasis is used in [16], an HSV colour space is used for feature extraction and the average accuracy achieved is 92%.

So, our study devises a deep learning framework specifically related to nail disease, with the acquired dataset of 4190 images.

III. DATA COLLECTION

Data Collection is the most important phase and is constitutes to be one of the biggest deep learning problems. The techniques in Deep Learning require a strong dataset for training the models. The better we train our model, more is the accuracy obtained after testing.

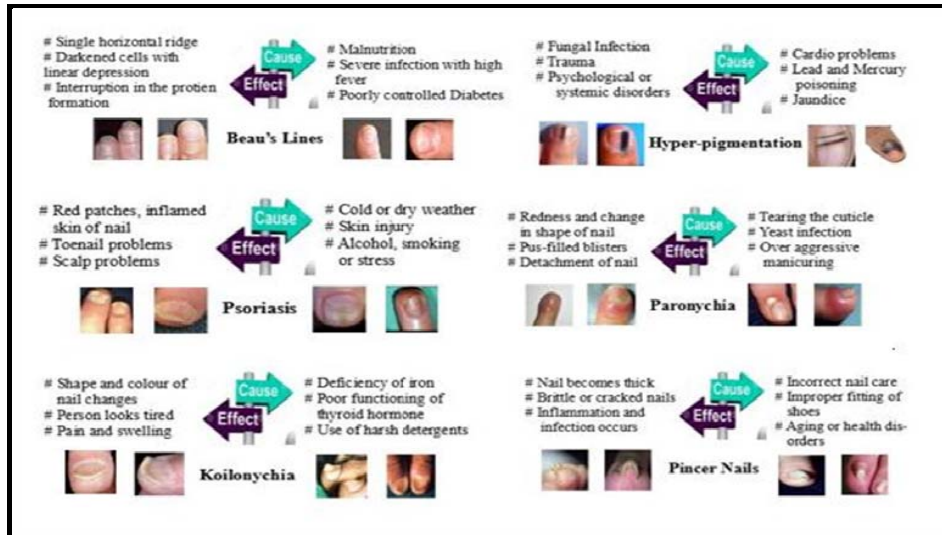
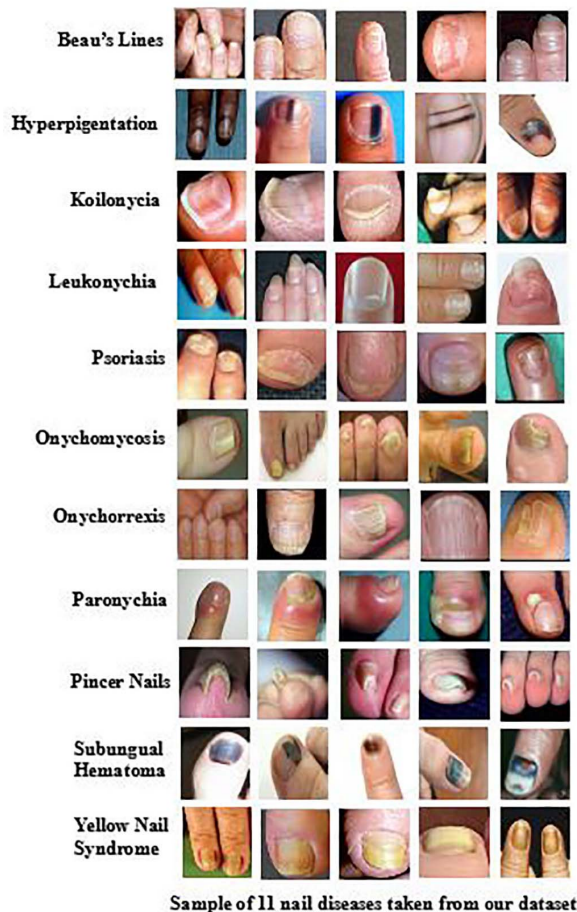


Figure 1: Nail Diseases



There are no open source datasets available for domains such as Nail Diseases with wide range of images that can be used to improve the state of art medical diagnosis. Due to this non-existence of the specific dataset for Nail Diseases, we created an image dataset, keeping a set of approximately 300 to 400 images, belonging to different disease classes (Fig. 2 and Table I). The causes and effects of the diseases are shown in Fig. 1.

We have chosen good quality images, with proper colour and dimensions, each representing the nail disease in question. We have also used AlexNet, which is a pre-trained model, trained on millions images, using the ImageNet dataset. We performed fine tuning on our dataset using the AlexNet architecture.

TABLE I. The table shows the total number of images present in the dataset corresponding to different Nail Diseases.

Disease Name	Number of Images
Beau's Lines	550
Hyperpigmentation	126
Koilonychia	174
Leukonychia	468
Psoriasis	350
Onychomycosis	482
Onychorrexis	302
Paronychia	422
Pincer Nails	457
Subungual Hematoma	403
Yellow Nail Syndrome	455
Total	4190

All the datasets were further divided into training and testing sets. For the datasets, 80% of the images in each category were randomly selected to make a training set and the rest were used to create a test set.

IV. METHODOLOGY

We used an ensemble of Convolutional Neural Network Architecture to learn the important features required for the classification of our class of nail diseases. The machine on which we have implemented our proposed constitutes a RAM of 8GB, a GPU of 8GB (Nvidia) with 280 cores, and an Intel i5 Processor.

First, we have performed fine tuning of the pre-trained alexnet architecture with our dataset. We also have run our dataset on some state of the art algorithms to find a contrast between our proposed approach and the accuracy obtained by the current state of the art algorithms (SVM [17], KNN [18], and RF [19]).

V. PROPOSED APPROACH

Our proposed approach uses deep Convolutional Neural Network to identify and classify a class of 11 nail diseases. We have done fine tuning of AlexNet architecture, which is

pre-trained on imageNet dataset, with our own dataset which is designed for nail diseases. The architecture of our model is based on [20], it consists of eight layers, constituting of five convolutional layers and three fully connected layers.

CNN has many hidden layers—generally varying from 5 to 7 in number. CNN has 60 million parameters, over six hundred thousand neurons at its service, and six hundred million connections. [21]

The standard way to model a neuron, determined by the following equation, i.e.

$$f(x) = \tanh(x) \text{ or } f(x) = (1 + e^{-x})^{-1} \quad \dots (1)$$

is a slow process. Therefore we use RELU, or “Non-saturating nonlinearity”, which is faster as compared. A CNN with RELU, converges six times faster than a network of $\tanh$ neurons, as claimed by [16]

$$f(x) = \max(0, x) \quad \dots (2)$$

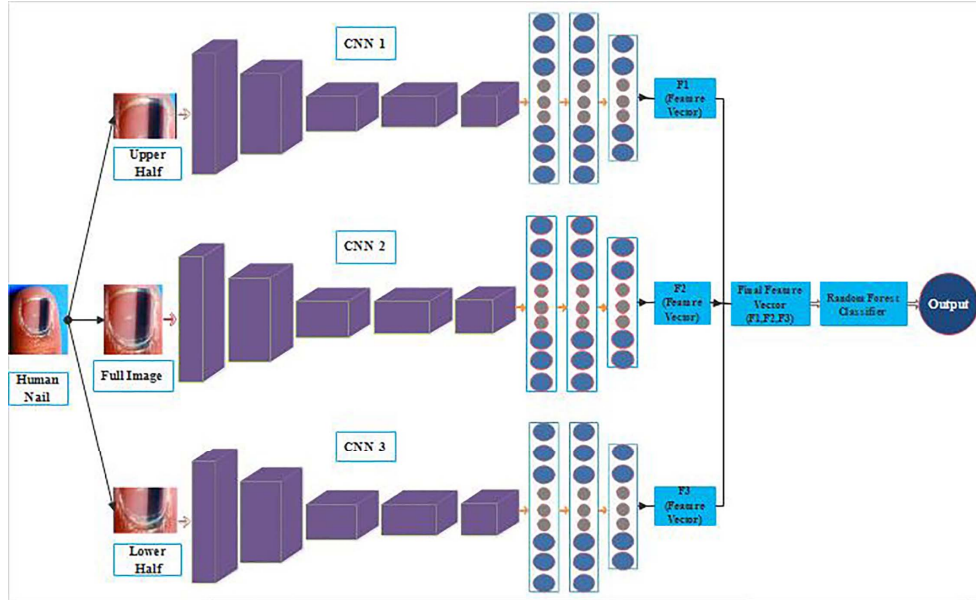


Figure 3: Our proposed architecture uses an ensemble of three such convolutional neural nets. We pass the full image to the first CNN A, the upper half of the image to the second CNN B and the lower half of the image to the third CNN C.

Using a hybrid of 3 different CNN's for capturing the different aspects of the human nail, the representations from the 7th fully connected layer of each network are extracted and stored in a vector, having dimension of 4096 x 1.

Three such vectors, obtained from each CNN were then combined together, forming the final feature vector, F such that $F = \{F1, F2, F3\}$ where $F1, F2$ and $F3$ are the obtained feature vectors. We apply Random Forest classifier on the final feature vector, F (Fig. 3).

We have also applied the current state of art algorithms such as Support Vector Machine (SVM), K-Nearest Neighbor, Random Forest, and CNN with SVM on our fine-tuned dataset and performed an accuracy assessment on the same.

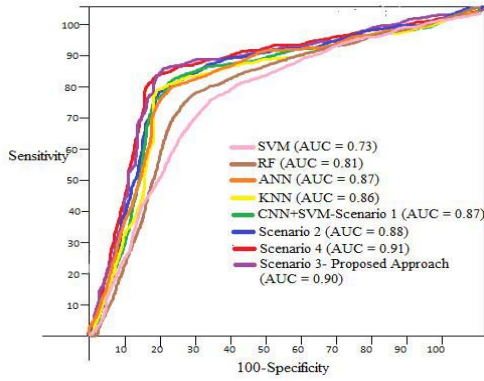


Figure 4: ROC Curves

We computed the ROC curves [22] for the evaluation of performance of our framework. It is built by plotting a “100-specificity” value on X-axis and “sensitivity” value on the Y axis which results into a graph. In our case, we can define sensitivity as the number of pixels that were correctly classified as a member of “disease” class [23]. The area under curve (AUC) represents the ROC curves. The closer the value of AUC is to 1, more is the degree of perfection of the framework [24].

VI. RESULTS AND DISCUSSIONS

This section brings a series of statistical plots with the finality of presenting the results obtained in the test proposed in sections above. We considered a total of four scenarios for accuracy assessment. It was observed that the accuracy of our proposed approach (Scenario 3) was the highest (84.58%) with a specificity of 0.88 and a sensitivity of 0.91. The performance of hybrid of CNNs is found to be comparable to practicing dermatologists and state of the art machine learning methods with only little pre-processing and fine-tuning. These results were achieved using networks pre-trained on other datasets with more or less related data. Fine tuning of the CNN architecture with our own dataset was also done and then the results were finally computed.

We considered a total of four different scenarios for accuracy assessment, taking into account all the cases. In scenario 1, we applied CNN on a full image of the nail. In scenario 2, we split an image into upper and lower halves respectively and supplied them simultaneous to two different. The feature vectors were combined to form one feature vector on which RF classification was performed. In scenario 3, we split the image into two parts again, and supplied the upper and lower halves to different CNNs, also supplying the full image to another CNN, combining the three feature vectors to form one, on which the final classification was applied using RF. In scenario 4, we split the image into four halves—the upper left, upper right, lower left and lower right, and supplied all four to four respective CNNs. However, during the accuracy assessment, it was observed that the difference in the accuracies of scenario 3(84.58%) and 4(85.11%) were comparable, but—the difference in the training time of scenario 4 was 4 hours more than that of scenario 3, hence taking the most optimal approach, we propose scenario 3.

It is seen that with the increase in convolutional layers in a framework, the accuracy increases due to precise and efficient feature extraction.

TABLE II. ACCURACY ASSESSMENT

Classification	Accuracy	Kappa Coefficient	Sensitivity	Specificity
SVM [17]	38.99%	0.287	0.434	0.382
RF[19]	76.57%	0.745	0.832	0.823
KNN[18]	77.19%	0.721	0.812	0.734
SCENARIO 1	78.21%	0.77	0.843	0.76
SCENARIO 2	80.45%	0.796	0.876	0.842
SCENARIO 3 (PA)	84.58%	0.841	0.91	0.88
SCENARIO 4	85.11%	0.848	0.898	0.891

VII. CONCLUSION

Our proposed approach used an ensemble of CNNs to classify 11 nail diseases, and achieved comparable accuracy of 80.45%. Due to the unavailability of a pre-existing database with substantial images on nail diseases, the task gained much more complexity, but it is imperative that there is scope for extension in the near future.

We aim to create a more generalised model that can be used for detection for other diseases such as hair diseases and teeth diseases.

Applications of our work may include an early detection of the nail disease which can get the required medical attention on time. Our work can also be extended and used by dermatologists to analyse the patient data and detect the disease with less manpower and human involvement.

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